CSE4261: Neural Network and Deep Learning
Assignment-5

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Answer

I selected the Macaw class from ImageNet, which corresponds to class label 88, and applied an adversarial attack using the Fast Gradient Sign Method (FGSM). Subsequently, I introduced Gaussian noise to the original image and re-applied the same adversarial attack to observe and compare the effects on model predictions.

Visual Comparison

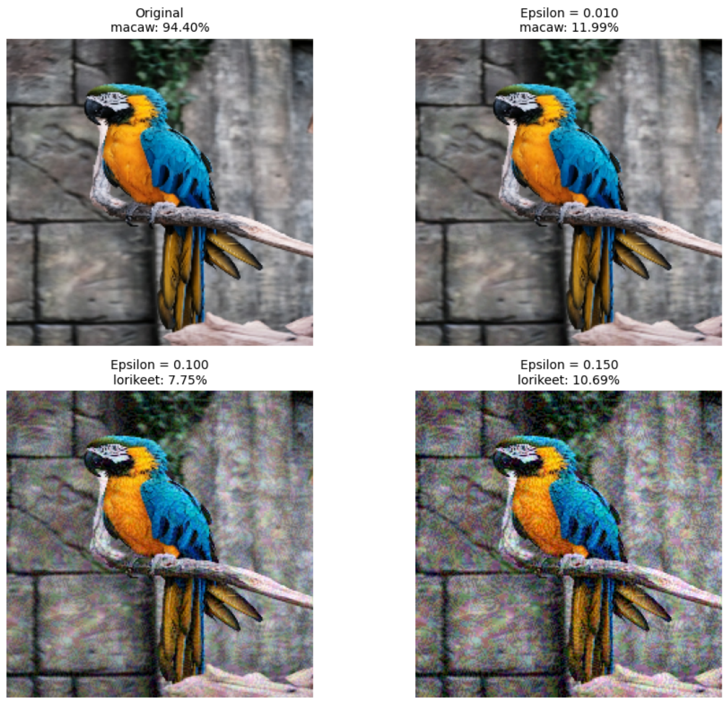

Figure 1: Adversarial attack on the original image (no Gaussian noise)

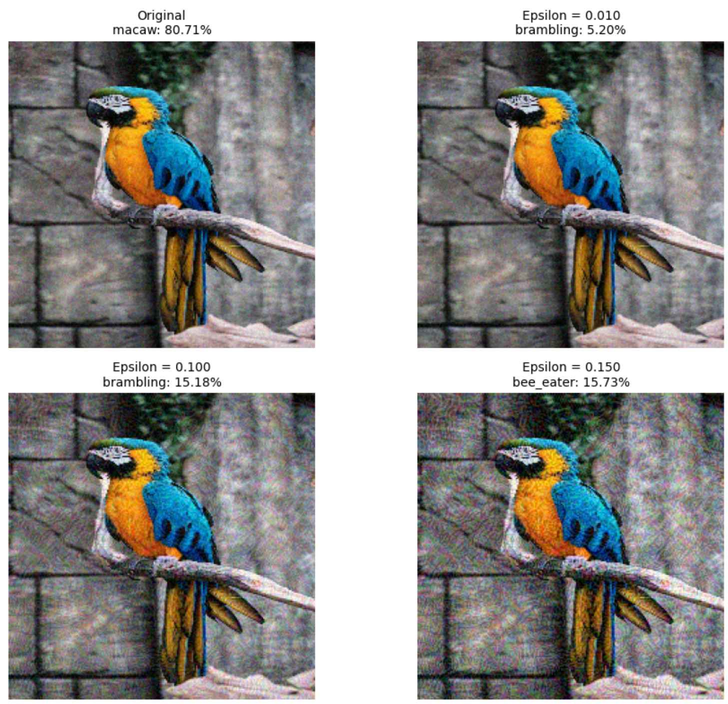

Figure 2: Adversarial attack on image after applying Gaussian noise

Comparison Table

Epsilon	Without Noise	With Gaussian Noise
0.000	Macaw: 94.40%	Macaw: 80.71%
0.010	Macaw: 11.99%	Brambling: 5.20%
0.100	Lorikeet: 7.75%	Brambling: 15.18%
0.150	Lorikeet: 10.69%	Bee-eater: 15.73%

Table 1: Model predictions under adversarial attack with different epsilon values

Discussion

The results indicate that adding Gaussian noise before applying an adversarial attack can amplify the vulnerability of the model. In the clean scenario (without noise), the image maintains a high classification confidence (94.40%) for the correct label *macaw*. After adding Gaussian noise, even before any adversarial perturbation, the confidence drops to 80.71%.
When small adversarial perturbations are added (e.g., $\epsilon = 0.01$), the model

with clean input still correctly classifies the image as *macaw*, albeit with low confidence (11.99%). However, after Gaussian noise is added, the same small perturbation already misclassifies the image as *brambling* with 5.20% confidence.

This demonstrates that Gaussian noise can act as a destabilizing factor. While Gaussian noise alone is random and non-targeted, it pushes the input closer to decision boundaries. When paired with gradient-based adversarial perturbations, the attack becomes more effective at smaller epsilon values.

Conclusion

Adding Gaussian noise does not by itself create adversarial examples but significantly weakens model confidence. When combined with adversarial attacks, Gaussian noise enhances the effectiveness of the attack. Therefore, such noise can be used as a subtle strategy to make models more susceptible to misclassification in adversarial settings.